

NGINX load balancer

index.js

```
const express = require("express")

const app = express()

app.get("/", (req, res) => {
  res.send("This is an endpoint");
})

app.listen(7777, () => {
  console.log("Listening on port 7777");
})
```

```
npm init -y
```

package.json (add start: node index)

```
{
  "name": "server",
  "version": "1.0.0",
  "description": "",
  "main": "index.js",
  "scripts": {
    "test": "echo \"Error: no test specified\" && exit 1",
    "start": "node index"
  },
  "keywords": [],
  "author": "",
  "license": "ISC",
  "dependencies": {
    "express": "^4.18.2"
  }
}
```

```
}  
}
```

test app:

```
npm run start
```

Dockerfile

```
# Fetching the minified node image on apline linux  
FROM node:slim  
  
# Setting up the work directory  
WORKDIR /express-docker  
  
# Copying all the files in our project  
COPY . .  
  
# Installing dependencies  
RUN npm install  
  
# Exposing server port  
EXPOSE 7777  
  
# Starting our application  
CMD [ "npm", "run", "start" ]
```

build containers:

```
docker build . -t myserver
```

create 4 instances of server:

```
sudo docker run -p 1111:7777 -d myserver
5d5de50c6f9a4f824897ec03e7cd3070f29a6a30fa364172c936d657c13f5
9e2
```

```
sudo docker run -p 2222:7777 -d myserver
ecbe41850007512ad87f197f74e172c14d3ed2a338e06210ad016ee535dde
9c0
```

```
sudo docker run -p 3333:7777 -d myserver
facf564c033a64f390ab8ac9740e5805af329acb7ad6d2b36dd8e65fe327d
349
```

```
sudo docker run -p 4444:7777 -d myserver
17cbbbbceb79093f6ead0d3e2b072c2b6ed78e5b32a57f180a0b21479c9085
1b6
```

nginx.conf:

```
http {

    include mime.types;

    upstream backendserver {
        server 127.0.0.1:1111;
        server 127.0.0.1:2222;
        server 127.0.0.1:3333;
        server 127.0.0.1:4444;
    }

    server {
        listen 80;
        root /home/pi/website;

        #regular expressions /count/9 works /count/10 is a
```

404

```
location ~* /count/[0-9]$ {
    root /home/pi/website;
    try_files /index.html = 404;
}

#load balancer in round robin mode
location / {
    proxy_pass http://backendserver/;
}

#/home/pi/website/hidden (looks for index.html)
location /hidden {
    root /home/pi/website;
}

#alias to another directory
location /another_hidden {
    alias /home/pi/website/hidden;
}

location /special {
    root /home/pi/website;
    #try look for
/home/pi/website/special/special_page.html if not
/home/pi/website/index.html
    try_files /special/special_page.html /index.html
=404;
}

location /move {
    #307 redirect to /hidden location
    return 307 /hidden;
}

#maintain url but rewrite to another dir
rewrite ^/number/([0-9])$ /count/$1;
}
```

```
}
```

```
events {}
```